Comparing Multi-Agent Coordination Rules via Fisher Information Loss after Action Aggregation

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Abstract

Multi-agent decision systems often separate *perception*—specialized agents that emit categorical distributions over local outcomes—from *coordination*—a deterministic rule that maps the joint outcome to a small action alphabet (e.g., reject, auto-apply, escalate to a human). While perception models and joint constructions have received careful treatment in Dec-POMDP active perception and information gathering, the coordination layer is frequently designed by heuristics and then evaluated only through task reward or throughput. This paper asks a complementary question: *holding the agents' output distributions fixed, which coordination rules destroy the least Fisher information about the underlying parameters?*

We formalize coordination as a θ-independent aggregation map $A = f(Z)$ and quantify information loss by the data-processing gap $I_Z(\theta) - I_A(\theta) \succeq 0$ between the Fisher information of the declared joint and that of its pushforward. Instantiating the framework in the Jevometry three-agent data-cleaning council (applicability, repair, risk) under a declared product joint, we compare six pure rules—baseline, stricter, wider, quarantine-to-review, coarse binary, and a near-identity upper bound—on a grid over evidence and ambiguity. At the design centre, baseline coordination discards roughly half of $\mathrm{tr}(I_Z)$; softening quarantine into human review recovers a substantial fraction of that loss, while coarsening to a binary reject/proceed gate loses still more. The ordering near-identity $\leq$ wider $\leq$ baseline $\leq$ stricter is stable at the centre and informative across the grid. We discuss design implications for agent councils, API constraints that force deterministic maps, and limitations of analytic softmax families relative to live decision-model APIs. Accompanying code and reproducible tables ship with the Jevometry repository.

1 Introduction

Large language models and specialized decision APIs have made it cheap to obtain calibrated-looking categorical distributions over local judgments: whether a repair applies, which repair to attempt, how risky an automatic write is. In production data systems these local judgments are rarely the end product. A coordinator—often a short piece of deterministic business logic—compresses the joint of agent outputs into a final action that can be executed, deferred, or rejected [Jevometry maintainers, 2026].

This compression is an information bottleneck. Classical decision theory and Dec-POMDP planning optimize rewards defined on beliefs or on prediction quality [Lauri and Oliehoek, 2020, Lauri et al., 2020, Oliehoek and Amato, 2016]. Information-theoretic multi-agent sensing maximizes mutual information or related set functions, often with submodular greedy guarantees [Krause et al., 2008, Nemhauser et al., 1978, Corah and Michael, 2021, Shi et al., 2026]. Those lines of work choose *what to observe* or *who should observe*. They do not, by themselves, tell a practitioner which of two already-deployed coordination rules retains more sensitivity to the parameters the agents were designed to track.

Fisher information provides a local, differentiable language for that sensitivity [Fisher, 1925, Amari, 2016]. For a discrete model $P_\theta(Z)$ with score Jacobian J_Z, the Fisher matrix $I_Z = J_Z^\top \mathrm{diag}(P_\theta)^{-1} J_Z$ measures how strongly infinitesimal parameter moves change the outcome distribution. When a fixed map f produces actions $A = f(Z)$, the data-processing inequality for Fisher information asserts $I_Z - I_A \succeq 0$ under standard regularity conditions [Zamir, 1998, Cover and Thomas, 2006]. The scalar $\mathrm{tr}(I_Z - I_A)$ is then a comparable loss summary when I_Z is held fixed.

Contributions.

1. We cast multi-agent coordination-rule comparison as ranking θ-independent aggregation maps by Fisher information loss under a shared declared joint (Section 3).
2. We instantiate six pure rules in the Jevometry cleaning council and provide a reproducible harness that sweeps a parameter grid, exports CSV/JSON, and emits publication figures (Section 4).
3. Empirically, at the design centre the baseline rule loses about 44% of $\mathrm{tr}(I_Z)$; a quarantine-to-review softening cuts loss by roughly a quarter relative to baseline; a binary gate loses more than stricter mid-risk rejection (Section 5).
4. We discuss how this metric complements reward-centric Dec-POMDP evaluation and what API constraints (deterministic maps only) imply for future learned coordinators (Section 6).

This manuscript is intended as a short journal / arXiv draft accompanying open-source experiments. It is not a claim about live cleaning quality; the agent families are analytic and illustrative.

2 Related work

Dec-POMDP active perception. Lauri and Oliehoek [2020] reduce multi-agent active perception with prediction rewards to standard Dec-POMDP solvers via individual prediction actions, with decentralization-loss bounds. Lauri et al. [2020] optimize convex belief-dependent rewards (e.g., negative entropy) with policy graph improvement in discrete and continuous-state settings. Both lines typically treat the team as fixed and fold information objectives into rewards rather than post-hoc comparing deterministic coordinators.

Information-driven multi-agent sensing. Submodular sensor selection and multi-robot tracking maximize mutual information or related coverage objectives with greedy $(1 - 1/e)$ guarantees under monotonicity [Krause et al., 2008, Corah and Michael, 2021]. Shi et al. [2026] jointly select agents and synthesize sensing policies for mutual-information objectives in Dec-POMDPs. Olkin et al. [2024] and Molloy and Nair [2023] further connect planning to information and smoother entropy. Our setting is dual: agents and their local laws are *fixed*; only the last-mile map to actions varies.

Information geometry for agents. Fisher–Rao geometry supplies local metrics on parametric families of distributions [Amari, 2016]. Jevometry operationalizes node- and system-level Fisher diagnostics for agent systems [Jevometry maintainers, 2026]. We use its pushforward `information_loss` primitive and do not modify the underlying DECLARED_PRODUCT construction.

3 Problem formulation

3.1 Declared product joint

Let $\theta \in \Theta \subset \mathbb{R}^d$ parameterize a multi-agent system. Each agent $i \in \{1, \ldots, m\}$ emits a categorical distribution $P_\theta(Z_i)$ over a finite local alphabet $\mathcal{Z}_i$. A *declared product joint* asserts conditional independence given θ:

$$P_\theta(Z = z) = \prod_{i=1}^{m} P_\theta(Z_i = z_i), \qquad z = (z_1, \ldots, z_m) \in \mathcal{Z} = \prod_i \mathcal{Z}_i. \tag{1}$$

In our cleaning example, $m = 3$, $d = 2$, and $\theta = (\text{evidence}, \text{ambiguity})$ with coordinates in $[0, 1]$. The declaration is an assumption of the model, not an empirical claim about live APIs.

3.2 Coordination as aggregation

Definition 1 (Coordination rule)**.** *A coordination rule is a map $f : \mathcal{Z} \to \mathcal{A}$ into a finite action alphabet $\mathcal{A}$. We call f* pure *if it is a total function of the joint outcome alone (no hidden randomness) and θ-*independent *if f does not depend on θ.*

The induced action law is the pushforward

$$P_\theta(A = a) = \sum_{z : f(z) = a} P_\theta(Z = z). \tag{2}$$

If $J_Z(\theta)$ is the Jacobian of the joint probability vector with respect to θ, the aggregate Jacobian J_A is obtained by summing rows of J_Z that share an action label.

3.3 Fisher information loss

Definition 2 (Information loss of a rule)**.** *For a θ-independent rule f, define*

$$I_Z(\theta) = J_Z(\theta)^\top \operatorname{diag}\big(P_\theta(Z)\big)^{-1} J_Z(\theta), \tag{3}$$

$$I_A(\theta) = J_A(\theta)^\top \operatorname{diag}\big(P_\theta(A)\big)^{-1} J_A(\theta), \tag{4}$$

$$\Delta_f(\theta) = I_Z(\theta) - I_A(\theta). \tag{5}$$

We report $\operatorname{tr}(I_Z)$, $\operatorname{tr}(I_A)$, and $\operatorname{tr}(\Delta_f)$ as scalar summaries, and we verify $\Delta_f \succeq 0$ numerically (PSD check) as a data-processing diagnostic.

Proposition 1 (DPI for fixed maps)**.** *Under standard positivity and differentiability conditions, any θ-independent f satisfies $\Delta_f(\theta) \succeq 0$. Equality holds when f is injective on the support of P_θ.*

Remark 1. *Maps that depend on θ (soft policies, learned routers conditioned on parameters) fall outside this guarantee in the Jevometry implementation and are reported as unsupported. This paper therefore restricts attention to pure, fully covering, θ-independent rules—the class that matches typical production if-then coordinators and the library's AggregationMap contract.*

Comparison principle. Fix the agents and the joint construction. Rank rules by $\operatorname{tr}(\Delta_f)$ (lower is better) or equivalently by $\operatorname{tr}(I_A)$ (higher is better). Because I_Z is shared, ranking by loss is identical to ranking by retained information.

4 Method and experimental design

4.1 Three-agent cleaning council

We reuse the offline Jevometry tutorial `examples/three_agent_cleaning` [Jevometry maintainers, 2026]. Three analytic softmax agents produce distributions for a single candidate repair of a noisy numeric field:

Agent	Primitive	Support
applicability	Noul	$\{\text{false}, \text{true}\}$
repair	Choice	$\{\text{normalize}, \text{impute}, \text{quarantine}\}$
risk	Score	$\{0, 1, 2\}$ (low/med/high)

Local logits are affine in $(\theta - \theta_0)$ around the design centre $\theta_0 = (0.7, 0.3)$, and probabilities are obtained by softmax. The joint is the declared product (ConstructionMode.DECLARED_PRODUCT). No live API is queried; no actual database write is performed.

4.2 Coordination rules

We compare six rules, all pure and θ-independent:

baseline Reject if inapplicable, quarantine, or high risk; auto-fix if normalize and low risk; otherwise human review.

stricter As baseline, but medium risk also rejects.

wider As baseline rejects; low-risk normalize and impute become distinct auto labels (`auto_normalize`, `auto_impute`).

quarantine_review Quarantine routes to human review instead of reject; reject only on inapplicable or high risk.

coarse_binary Reject under baseline reject conditions; otherwise a single `proceed` action (auto-fix and review collapsed).

near_identity One action label per joint outcome tuple (18 labels): Fisher retention upper bound under this alphabet.

4.3 Harness

The script `paper_experiments.py` builds nodes and joint once, evaluates I_Z, I_A, and Δ_f on an 8×8 grid over evidence $\in [0.2, 0.9]$ and ambiguity $\in [0.1, 0.8]$, and additionally records the exact centre θ_0. Outputs:

- `papers/coord-fisher/data/rule_loss_grid.{csv,json}`
- `papers/coord-fisher/data/center_summary.json`
- Figures 1–4 under `papers/coord-fisher/figures/`

Library code under `src/jevometry` is unchanged; only example-level rules and scripts are extended.

Table 1: Fisher traces at centre $\theta_0 = (0.7, 0.3)$. $|A|$ is the number of actions with positive mass under the pushforward.

Rule	$\mathrm{tr}(I_Z)$	$\mathrm{tr}(I_A)$	$\mathrm{tr}(\Delta)$	$\lvert A\rvert$	Status
near_identity	6.8418	6.8418	0.0000	18	ok
quarantine_review	6.8418	4.5764	2.2654	3	ok
wider	6.8418	3.8285	3.0133	4	ok
baseline	6.8418	3.8080	3.0338	3	ok
stricter	6.8418	3.4198	3.4220	3	ok
coarse_binary	6.8418	3.1296	3.7122	2	ok

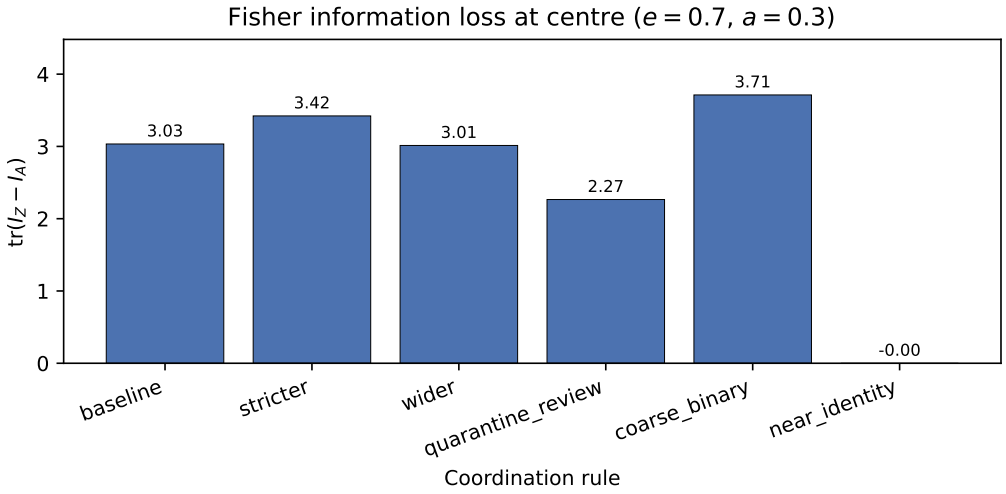


Figure 1: Information loss $\mathrm{tr}(I_Z - I_A)$ by rule at the design centre.

5 Experiments

5.1 Centre-point ranking

Table 1 reports traces at θ_0. All rules return status `ok` with PSD loss matrices. The near-identity map loses nothing (up to numerical noise), confirming the implementation against the equality case of the DPI. Baseline loses $\mathrm{tr}(\Delta) \approx 3.03$ out of $\mathrm{tr}(I_Z) \approx 6.84$—about 44%. Wider slightly improves on baseline by preserving distinct auto paths. Stricter mid-risk rejection increases loss. Softening quarantine into review yields the largest practical gain among coarse three/four-action policies ($\mathrm{tr}(\Delta) \approx 2.27$). Collapsing to a binary gate is worst among the listed rules (≈ 3.71).

The core ordering highlighted in prior exploratory work remains:

$$\text{near-identity} \;\leq\; \text{wider} \;\leq\; \text{baseline} \;\leq\; \text{stricter}$$

at the centre (numerical tolerances 10^{-9}). Quarantine-to-review inserts *below* wider on the loss axis, and coarse binary sits *above* stricter—both are informative deviations from the original four-rule story.

5.2 Grid behaviour

Figure 2 shows loss heatmaps for five non-identity rules over the parameter grid. Absolute loss varies with θ because both $I_Z(\theta)$ and the mass allocation across f-fibres change, but the qualitative ranking is stable in large regions of the domain: softer, finer action alphabets sit cooler (lower loss) than stricter or binary gates. Figure 3 slices loss against evidence at ambiguity nearest 0.3; near-identity stays at zero, while coarse binary and stricter remain the most lossy.

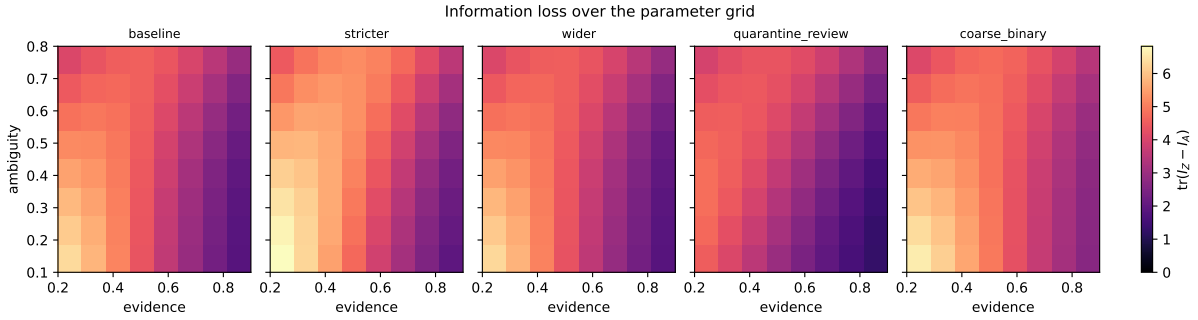


Figure 2: Loss heatmaps over evidence×ambiguity for five coordination rules.

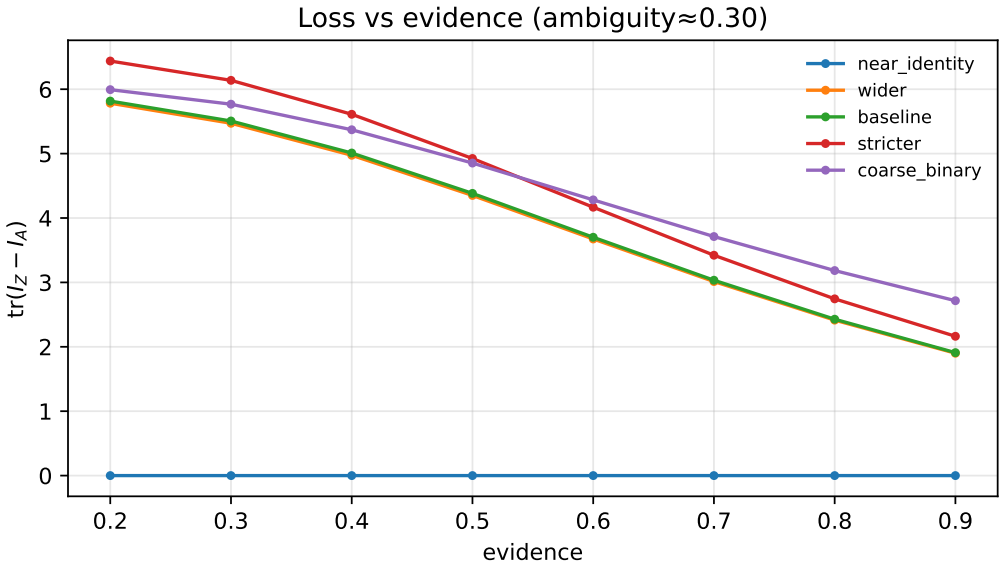


Figure 3: Loss versus evidence at fixed ambiguity near 0.3.

5.3 Action masses

Figure 4 shows how probability mass is partitioned at the centre. Stricter reallocates medium-risk mass into reject. Wider splits auto-apply into normalize versus impute. Quarantine-to-review shrinks reject in favour of human review. Coarse binary folds auto-fix and review into `proceed`. These mass shifts are the mechanism behind Fisher loss: merging outcomes with dissimilar score rows collapses distinguishable parameter directions.

6 Discussion

What "half the Fisher" means. A 44% drop in $\mathrm{tr}(I_Z)$ after baseline coordination does not mean the system is 44% "wrong." It means that, locally, infinitesimal changes in evidence and ambiguity move the final action distribution less than they move the joint of agent outputs. If a monitoring dashboard only watches action rates, it understates how sensitive the underlying council is. If a downstream estimator tries to invert actions for θ, it faces a strictly harder problem than inverting Z.

Design takeaway. Among implementable three/four-action policies, routing quarantine to review retained the most Fisher in this model, beating both the historical baseline and the wider auto-impute split. Binary proceed/reject gates are convenient operationally but costly

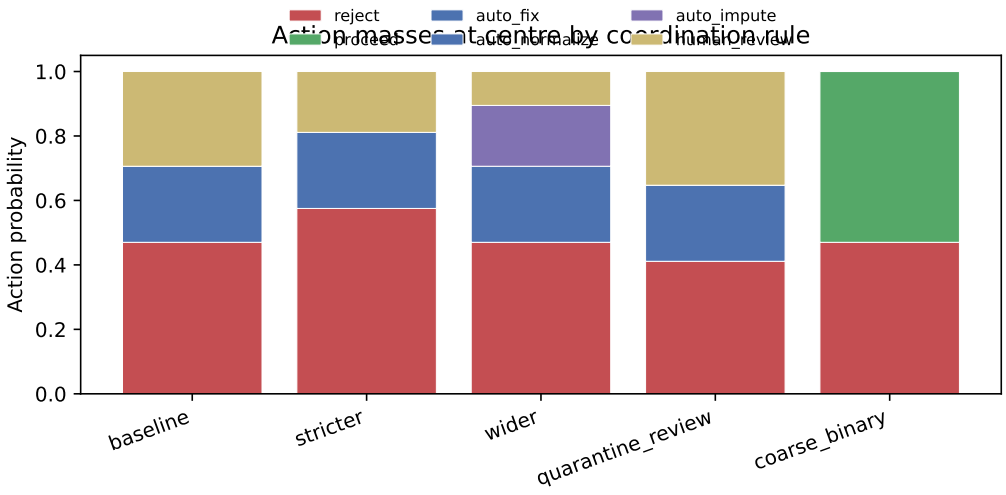


Figure 4: Pushforward action probabilities at the centre for non-identity rules.

informationally. Practitioners can treat $\mathrm{tr}(\Delta_f)$ as a cheap offline score when debating rule changes before A/B tests.

Relation to Dec-POMDP rewards. Prediction rewards and negative-entropy objectives [Lauri and Oliehoek, 2020, Lauri et al., 2020] optimize perception policies. Our metric does not replace them; it audits the *last* deterministic compression after perception is fixed. In that sense it is complementary to IMAS^2-style joint selection [Shi et al., 2026]: selection expands who may observe; coordination ranking asks how much of that observation survives the action layer.

API constraints. Many agent platforms expose only sampled outcomes plus deterministic business rules. Soft, θ-dependent routers would require either a declared conditional action model or a refusal of the DPI certificate—exactly the contract encoded by Jevometry's AggregationMap. Extending the present study to stochastic coordinators is future work and needs additional regularity.

7 Limitations

- Agent distributions are hand-specified analytic softmax families, not fitted live decision models. Absolute numbers are illustrative.
- Conditional independence is declared, not tested.
- Trace summaries ignore anisotropic structure in I_Z and I_A; full matrix comparisons or parameter-wise CRLB-style reads would need an observation likelihood contract this example does not provide.
- Action alphabets differ across rules; $\mathrm{tr}(I_A)$ comparisons remain meaningful as retained information about θ, but interpretations that treat $|A|$ as utility are out of scope.
- No human or task reward study validates that lower Fisher loss improves cleaning outcomes.

8 Conclusion

We proposed ranking multi-agent coordination rules by Fisher information loss under a shared declared joint and demonstrated the approach on a reproducible three-agent cleaning council. Holding perception fixed, deterministic coordinators can discard on the order of half the joint Fisher trace; small rule edits (quarantine handling, auto-path splitting, risk thresholds) move that loss by amounts that are easy to measure offline. The accompanying harness and figures are intended as a template for auditing production if-then councils before changing live traffic.

Future work includes (i) replacing analytic agents with TypeSafe / live decision-model Jacobians under the same AggregationMap interface, (ii) stochastic coordinators with explicit conditional action models, and (iii) coupling $\mathrm{tr}(\Delta_f)$ with task-level utilities in JAAMAS-style empirical studies.

Reproducibility

From the repository root:

```
uv sync --locked
uv pip install matplotlib   # paper figures
uv run python examples/three_agent_cleaning/compare_rules.py
uv run python examples/three_agent_cleaning/paper_experiments.py
```

Artefacts land in `papers/coord-fisher/data` and `figures/`. This PDF was built with Tectonic from `papers/coord-fisher/main.tex`.

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